

Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(a)	Deduces that $A = \pm 50$ or $b = \frac{1}{4}$	B1	3.4
	Deduces that $A = \pm 50$ and $b = \frac{1}{4}$	B1	3.4
	Uses $t = 0, H = 1 \Rightarrow \alpha = \dots$ E.g. $1 = "50" \sin(\alpha)^\circ \Rightarrow \alpha = \dots$	M1	3.4
	$H = \left \pm 50 \sin\left(\frac{1}{4}t + 1.15\right)^\circ \right $	A1	3.3
		(4)	
(b)	E.g. the minimum height above the ground of the passenger on the original model was 0 m or Adding "d" means the passenger does not touch the ground.	B1	3.5b
		(1)	
(5 marks)			
Notes:			

(a) Note that B0B1 is not possible

B1: Uses the equation of the given model to deduce that $A = \pm 50$ or $b = \frac{1}{4}$ o.e.

May be seen embedded within their equation.

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May be seen embedded within their equation.

M1: Uses $t = 0$ and $H = 1$ in the equation of the model to find a value for α .

Follow through on their value for A . Allow for $\pm 1 = "50" \sin(\alpha)^\circ \Rightarrow \alpha = \dots$ where α is in degrees or radians.

Note that in radians $\sin^{-1}\left(\frac{1}{50}\right) \approx \frac{1}{50}$ (0.0200...) which may appear incorrect but is in fact ok.

Also in degrees a value of e.g. 1.14 (truncated) would indicate the method.

A1: Writes down the correct full equation of the model: $H = \left| \pm "50" \sin\left(\frac{1}{4}t + 1.15\right)^\circ \right|$ o.e.

Condone omission of degrees symbol and allow awrt 1.15 for α .

Allow if a correct equation is seen anywhere in their solution.

(b)

B1: Gives a suitable explanation with no contradictory statements.

Condone "so that pod/capsule/seat/passenger/ferris wheel/it etc. will not hit/touch the ground"

Responses that focus on the starting point of the model are likely to score B0

Q2.

Question	Scheme	Marks	AOs
(a)	Attempts to use $h^2 = at + b$ with either $t = 2, h = 2.6$ or $t = 10, h = 5.1$	M1	3.1b
	Correct equations $2a + b = 6.76$ $10a + b = 26.01$	A1	1.1b
	Solves simultaneously to find values for a and b	dM1	1.1b
	$h^2 = 2.41t + 1.95$ cao	A1	3.3
		(4)	
(b)	Substitutes $t = 20$ into their $h^2 = 2.41t + 1.95$ and finds h or h^2 Or substitutes $h = 7$ into their $h^2 = 2.41t + 1.95$ and finds t	M1	3.4
	Compares the model with the true values and concludes "good model" with a minimal reason E.g. I Finds $h = 7.08$ (m) and states that it is a good model as 7.08 (m) is close to 7 (m) E.g. II Finds $t = 19.5$ years and states that the model is accurate as 19.5 (years) $\approx$ 20 (years)	A1	3.5a
		(2)	
(6 marks)			
Notes:			

(a)

M1: For translating the problem into mathematics. Attempts to use the given equation o.e. with either of the pieces of information to form one correct equation.

Award for unsimplified equations as well, such as $2.6^2 = 2a + b$ or $2.6 = \sqrt{2a + b}$

A1: Two correct (and different) equations which may be unsimplified

dM1: Solves simultaneously to find values for a and b . It is dependent upon the previous M

Don't be too concerned with the process here as calculators may be used.

Score if values of a and b are reached from a pair of simultaneous equations

A1: Establishes the full equation of the model with values of a and b given to exactly 3sf. Award if seen in either (a) or (b). It is not scored for the values of a and b .

Allow either $h^2 = 2.41t + 1.95$ or $h = \sqrt{2.41t + 1.95}$

If they go on to square root each term from $h^2 = 2.41t + 1.95$ then it is A0. E.g. $h = 1.55t + 1.40$

Special case for candidates who mistakenly use $h = at + b$

For $2.6 = 2a + b$, $5.1 = 10a + b \Rightarrow h = 0.3125t + 1.975$ or $h = 0.313t + 1.98$

can score M1 correct equations with attempt to solve and A1 for either correct answer shown above.

These are the only marks available to them for a maximum mark of 1100 00

(b)

M1: A full and valid attempt to

either substitute $t = 20$ into their $h^2 = 2.41t + 1.95$ o.e. and find a value for h or h^2

or substitute $h = 7$ into their $h^2 = 2.41t + 1.95$ o.e. and find a value for t

(to enable the candidate to compare real life data with that of the model.)

The equation of the model must be of the correct form, either $h^2 = at + b$ or $h = \sqrt{at + b}$

Do not be too concerned with the mechanics of the solution but the square or $\sqrt{\quad}$ must have been used appropriately to enable the comparison to be made.

In cases with no working you will need to check the calculation

A1: Compares their $h = 7.08\text{m}$ to 7m o.e using h^2 or their $t = 19.5$ years to 20 years and makes valid conclusion with reason.

For this mark you require

- a statement that it is a "good" or "accurate" model or similar wording
- a reason such as "the values are close", "the values are similar" or "the predicted values are within 5% of the true values."
- a model with equation $h^2 = at + b$ o.e. where $a = \text{awrt } 2.4$ and $b \in [1.9, 2.0]$
- correct calculations

Condone a statement like 'the model is pretty accurate as it predicted 7.08m and the actual value is 7m'

Do not allow incorrect statements such as the model is incorrect as it does not give 7 metres.

Do not allow just "the model gives an underestimate of the true value."

Do not allow 'bad' or 'poor' model

Q3.

Question	Scheme	Marks	AOs
(a) Way 1	$H = Ax(40 - x)$ {or $H = Ax(x - 40)$ }	M1	3.3
	$x = 20, H = 12 \Rightarrow 12 = A(20)(40 - 20) \Rightarrow A = \frac{3}{100}$	dM1	3.1b
	$H = \frac{3}{100}x(40 - x)$ or $H = -\frac{3}{100}x(x - 40)$	A1	1.1b
	(3)		
(a) Way 2	$H = 12 - \lambda(x - 20)^2$ {or $H = 12 + \lambda(x - 20)^2$ }	M1	3.3
	$x = 40, H = 0 \Rightarrow 0 = 12 - \lambda(40 - 20)^2 \Rightarrow \lambda = \frac{3}{100}$	dM1	3.1b
	$H = 12 - \frac{3}{100}(x - 20)^2$	A1	1.1b
	(3)		
(a) Way 3	$H = ax^2 + bx + c$ (or deduces $H = ax^2 + bx$) Both $x = 0, H = 0 \Rightarrow 0 = 0 + 0 + c \Rightarrow c = 0$ and either $x = 40, H = 0 \Rightarrow 0 = 1600a + 40b$ or $x = 20, H = 12 \Rightarrow 12 = 400a + 20b$ or $\frac{-b}{2a} = 20 \Rightarrow b = -40a$	M1	3.3
	$b = -40a \Rightarrow 12 = 400a + 20(-40a) \Rightarrow a = -0.03$ so $b = -40(-0.03) = 1.2$	dM1	3.1b
	$H = -0.03x^2 + 1.2x$	A1	1.1b
	(3)		
(b)	$\{H = 3 \Rightarrow\} 3 = \frac{3}{100}x(40 - x) \Rightarrow x^2 - 40x + 100 = 0$ or $\{H = 3 \Rightarrow\} 3 = 12 - \frac{3}{100}(x - 20)^2 \Rightarrow (x - 20)^2 = 300$	M1	3.4
	e.g. $x = \frac{40 \pm \sqrt{1600 - 4(1)(100)}}{2(1)}$ or $x = 20 \pm \sqrt{300}$	dM1	1.1b
	{chooses $20 + \sqrt{300} \Rightarrow$ } greatest distance = awrt 37.3 m	A1	3.2a
	(3)		
(c)	Gives a limitation of the model. Accept e.g. - the ground is horizontal - the ball needs to be kicked from the ground - the ball is modelled as a particle - the horizontal bar needs to be modelled as a line - there is no wind or air resistance on the ball - there is no spin on the ball - no obstacles in the trajectory (or path) of the ball - the trajectory of the ball is a perfect parabola	B1	3.5b
	(1)		
(7 marks)			

Notes for Question	
(a)	
M1:	Translates the situation given into a suitable equation for the model. E.g. Way 1: {Uses (0, 0) and (40, 0) to write} $H = Ax(40 - x)$ o.e. {or $H = Ax(x - 40)$} Way 2: {Uses (20, 12) to write} $H = 12 - \lambda(x - 20)^2$ or $H = 12 + \lambda(x - 20)^2$ Way 3: Writes $H = ax^2 + bx + c$, and uses (0, 0) to deduce $c = 0$ and an attempt at using either (40, 0) or (20, 12) Special Case: Allow SC M1dM0A0 for not deducing $c = 0$ but attempting to apply both (40, 0) and (20, 12)
dM1:	Applies a complete strategy with appropriate constraints to find all constants in their model. Way 1: Uses (20, 12) on their model and finds $A = \dots$ Way 2: Uses either (40, 0) or (0, 0) on their model to find $\lambda = \dots$ Way 3: Uses (40, 0) and (20, 12) on their model to find $a = \dots$ and $b = \dots$
A1:	Finds a correct equation linking H to x E.g. $H = \frac{3}{100}x(40 - x)$, $H = 12 - \frac{3}{100}(x - 20)^2$ or $H = -0.03x^2 + 1.2x$
Note:	Condone writing y in place of H for the M1 and dM1 marks.
Note:	Give final A0 for $y = -0.03x^2 + 1.2x$
Note:	Give special case M1dM0A0 for writing down any of $H = 12 - (x - 20)^2$ or $H = x(40 - x)$ or $H = x(x - 40)$
Note:	Give M1 dM1 for finding $-0.03x^2 + 1.2x$ or $a = -0.03, b = 1.2, c = 0$ in an implied $ax^2 + bx + c$ or $ax^2 + bx + c$ (with no indication of $H = \dots$)
(b)	
M1:	Substitutes $H = 3$ into their quadratic equation and proceeds to obtain a 3TQ or a quadratic in the form $(x \pm \alpha)^2 = \beta; \alpha, \beta \neq 0$
Note:	E.g. $1.2x - 0.03x^2 = 3$ or $40x - x^2 = 100$ are acceptable for the 1 st M mark
Note:	Give M0 dM0 A0 for $(\text{their } A)x^2 = 3 \Rightarrow x = \dots$ or their $(\text{their } A)x^2 + (\text{their } k) = 3 \Rightarrow x = \dots$
dM1:	Correct method of solving their quadratic equation to give <i>at least one solution</i>
A1:	Interprets their solution in the original context by selecting the larger correct value <i>and states correct units for their value</i>. E.g. Accept awrt 37.3 m or $(20 + \sqrt{300})$ m or $(20 + 10\sqrt{3})$ m
Note:	Condone the use of inequalities for the method marks in part (b)
(c):	
B1:	See scheme
Note:	Give no credit for the following reasons H (or the height of ball) is negative when $x > 40$ - Bounce of the ball should be considered after hitting the ground - Model will not be true for a different rugby ball - Ball may not be kicked in the same way each time

Q4.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$6 = \pi r^2 h + \frac{2}{3} \pi r^3$	B1	This mark is given for a method to find the volume of the cylinder and the semi-hemisphere
	$A = 3\pi r^2 + 2\pi \left(\frac{6 - \frac{2}{3}\pi r^3}{\pi r} \right)$	M1	This mark is given for a method to find the surface area of the tank
		A1	This mark is given for finding an expression for the surface area of the tank
	$A = 3\pi r^2 + \frac{12}{r} - \frac{4\pi r^2}{3} = \frac{12}{r} + \frac{5\pi r^2}{3}$	A1	This mark is given for a fully correct proof to show the surface area of the tank as required
(b)	$A = \frac{12}{r} + \frac{5\pi r^2}{3} \Rightarrow \frac{dA}{dr} = -\frac{12}{r^2} + \frac{10\pi r}{3}$	M1	This mark is given for a method to differentiate to find r
		A1	This mark is given for accurately differentiating to find r
	When $\frac{dA}{dr} = 0$, $-\frac{12}{r^2} + \frac{10\pi r}{3} = 0$ $r^3 = \frac{18}{5\pi}$	M1	This mark is given for a method to set $\frac{dA}{dr} = 0$ to find a value for r
	$r = 1.046$	A1	This mark is given for finding the radius for which the surface area is a minimum
(c)	$A = \frac{12}{1.046} + \frac{5\pi(1.046)^2}{3}$	M1	This mark is given for a method to substitute a value for r
	$A = 17 \text{ m}^2$	A1	This mark is given for correctly finding the minimum surface area of the tank (to the nearest integer)
(Total 10 marks)			

Q5.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$y = a + kx$, where a and k are constants	B1	This mark is given for stating a correct general equation
(b)	$500 = 800 \times 2 - (a + 800k)$ $-80 = 300 \times 2 - (a + 300k)$	M1	This mark is given for modelling the profit on the two days when bars of soap are sold for £2
	$a + 800k = 1100$ $a + 300k = 680$	M1	This mark is given for forming a pair of simultaneous equations to find values for a and k
	$500k = 420 \Rightarrow k = \frac{420}{500} = 0.84$ $a + (800 \times 0.84) = 1100$ $a = 1100 - 672 = 428$ Thus $y = 0.84x + 428$	A1	This mark is given for finding the values of a and k to show $y = 0.84x + 428$ as required
(c)	0.84 represents the cost of making one extra bar of soap in £s (i.e. 84p)	B1	This mark is given for a valid interpretation of the significance of 0.84
(d)	For n bars of soap $2n - (428 + 0.84n) > 0$	M1	This mark is given for a method to find the number of bars of soap to be made
	$1.16n - 428 > 0$ $n - \frac{428}{1.16} > 0$ $n = 369$ bars of soap	A1	This mark is given for correctly finding the number of bars of soap to be made
(Total 7 marks)			

Q6.

Question	Scheme	Marks	AOs
(a)	265 thousand	B1	3.4
		(1)	
(b)	Attempts $\frac{dN_b}{dt} = 11e^{0.05t}$	M1	1.1b
	Substitutes $t = 10$ into their $\frac{dN_b}{dt}$	M1	3.4
	$\frac{dN_b}{dt} = \text{awrt } 18.1$ which is approximately 18 thousand per year *	A1*	2.1
		(3)	
(c)	Sets $45 + 220e^{0.05t} = 10 + 800e^{-0.05t} \Rightarrow 220e^{0.05t} + 35 - 800e^{-0.05t} = 0$	M1	3.1b
	Correct quadratic equation $\Rightarrow 220(e^{0.05t})^2 + 35e^{0.05t} - 800 = 0$	A1	1.1b
	$e^{0.05t} = 1.829, (-1.988) \Rightarrow 0.05t = \ln(1.829)$	M1	2.1
	$T = 12.08$	A1	1.1b
		(4)	
(8 marks)			
Notes:			

(a) May be seen in the question so watch out.

B1: Accept 265 thousand or 265 000 or equivalent such as 265 k but not just 265.

(b)

M1: Differentiates to a form $ke^{0.05t}$, $k > 0, k \neq 220$. Do not be too concerned about the lhs.

M1: Substitutes $t = 10$ into a changed function that was formed from an attempt at differentiation.

The left hand side must have implied differentiation. E.g. Rate = , N' , $\frac{dN_b}{dt}$, $\frac{dN}{dt}$ or even $\frac{dy}{dx}$

A1*: Full and complete proof that requires

- some correct lhs seen at some point. E.g. "Rate = , " $\frac{dN_b}{dt}$, $\frac{dN}{dt}$ but condone N' .
- an intermediate line/answer of either $11e^{0.05 \times 10}$ or awrt 18.1 before a minimal conclusion which must be referencing the 18 000 or 18 thousand

(c)

M1: Attempts to set both equations equal to each other and simplify the constant terms.

Look for $220e^{0.05t} + 35 = 800e^{-0.05t}$ o.e but condone slips

It is also possible to set $\frac{N-45}{220} = \left(e^{0.05t} = \frac{800}{N-10}\right)$ and form an equation in N

A1: Correct quadratic form.

Look for $220(e^{0.05t})^2 + 35e^{0.05t} - 800 = 0$ or $220e^{0.1t} + 35e^{0.05t} - 800 = 0$ but allow with terms in different order such as $220e^{0.1t} + 35e^{0.05t} = 800$

FYI the equation in N is $N^2 - 55N - 175550 = 0$

M1: Full attempt to find the value of t (or a constant multiple of t)

This involves the key step of recognising and solving a 3TQ in $e^{0.05t}$ followed by the use of lns.

If the answers to the quadratic just appear (from a calculator) you will need to check.

Accuracy should be to 3sf.

You may see different variables used such as x

$$x = e^{0.05t}, 220e^{0.1t} + 35e^{0.05t} = 800 \Rightarrow 220x^2 + 35x = 800 \Rightarrow x = 1.82... \Rightarrow t = 20 \ln 1.82...$$

Allow use of calculator for solving the quadratic and for $e^{0.05t} = 1.82... \Rightarrow t = 12.08$

Via the N route it will involve substituting the positive solution to their quadratic into either equation to find a value for t/T using same rules as above.

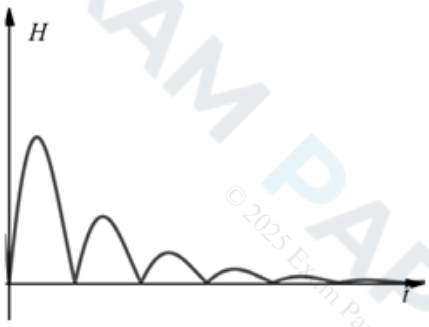
A1: AWRT 12.08

Answers with limited or no working in (b) and (c)

(b) A derivative in the correct form must be seen

(c) Candidates who state $45 + 220e^{0.05t} = 10 + 800e^{-0.05t}$ followed by awrt 12.08 (presumably from using num-solv on their calculators) can score SC 1100. Rubric on the front of the paper states that "Answers without working may not gain full credit" so we demand a method in this part.

Q7.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$f'(x) = -2.5e^{-0.25x} \sin x + 10e^{-0.25x} \cos x$	M1	This mark is given for a method to differentiate to find an expression for $f'(x)$
		A1	This mark is given for correctly differentiating to find an expression for $f'(x)$
	$f'(x) = 0$ $\Rightarrow e^{-0.25x} (-2.5 \sin x + 10 \cos x) = 0$ $\Rightarrow (-2.5 \sin x + 10 \cos x) = 0$	M1	This mark is given for setting $f'(x) = 0$ and finding a method to solve for $\tan x$
	$\frac{\sin x}{\cos x} = \frac{10}{2.5}$ $\tan x = 4$	A1	This mark is given for showing that $\tan x = 4$ as required.
(b)		M1	This mark is given for a graph with a correct shape
		A1	This mark is given for a graph with heights > 0
(c)	$\tan x = 4, x = 1.326$ $t = \pi + 1.326 = 4.47$	M1	This method is given for finding a value for t between the first and second bounce
	$H(4.47) = 10e^{-0.25 \times 4.47} \sin 4.47 $	M1	This mark is given for substituting the value of $t = \pi + \arctan 4$ into $H(t)$
	$= 3.27 \times -0.97 $ $= 3.17 \text{ metres}$	A1	This mark is given for finding the maximum height of the ball
(d)	The time between each bounce should not stay the same when the heights of each bounce are getting smaller	B1	This mark is given for a valid explanation of why the model should not be used to predict the time of each bounce

Q8.

Question	Scheme	Marks	AOs
(a)(i)	10750 barrels	B1	3.4
(ii)	Gives a valid limitation, for example - The model shows that the daily volume of oil extracted would become negative as t increases, which is impossible - States when $t = 10, V = -1500$ which is impossible - States that the model will only work for $0 \leq t \leq \frac{64}{7}$	B1	3.5b
		(2)	
(b)(i)	Suggests a suitable exponential model, for example $V = Ae^{kt}$	M1	3.3
	Uses (0,16000) and (4,9000) in $\Rightarrow 9000 = 16000e^{4k}$	dM1	3.1b
	$\Rightarrow k = \frac{1}{4} \ln\left(\frac{9}{16}\right)$ awrt -0.144	M1	1.1b
	$V = 16000e^{\frac{1}{4} \ln\left(\frac{9}{16}\right)t}$ or $V = 16000e^{-0.144t}$	A1	1.1b
(ii)	Uses their exponential model with $t = 3 \Rightarrow V =$ awrt 10 400 barrels	B1ft	3.4
		(5)	
(7 marks)			

Notes:

(a)(i)

B1: 10750 barrels

(a)(ii)

B1: See scheme

(b)(i)

M1: Suggests a suitable exponential model, for example $V = Ae^{kt}$, $V = Ar^t$ or any other suitable function such as $V = Ae^{kt} + b$ where the candidate chooses a value for b .

dM1: Uses both (0,16000) and (4,9000) in their model.

With $V = Ae^{kt}$ candidates need to proceed to $9000 = 16000e^{4k}$

With $V = Ar^t$ candidates need to proceed to $9000 = 16000r^4$

With $V = Ae^{kt} + b$ candidates need to proceed to $9000 = (16000 - b)e^{4k} + b$ where b is given as a positive constant and $A + b = 16000$.

M1: Uses a correct method to find all constants in the model.

A1: Gives a suitable equation for the model passing through (or approximately through in the case of decimal equivalents) both values (0,16000) and (4,9000). Possible equations for the model could be for example

$$V = 16000e^{-0.144t} \quad V = 16000 \times (0.866)^t \quad V = 15800e^{-0.146t} + 200$$

(b)(ii)

B1ft: Follow through on their exponential model

Question	Scheme	Marks	AOs
(a)	Sets $H = 0 \Rightarrow 1.8 + 0.4d - 0.002d^2 = 0$	M1	3.4
	Solves using an appropriate method, for example $d = \frac{-0.4 \pm \sqrt{(0.4)^2 - 4(-0.002)(1.8)}}{2 \times -0.002}$	dM1	1.1b
	Distance = awrt 204(m) only	A1	2.2a
		(3)	
(b)	States the initial height of the arrow above the ground.	B1	3.4
		(1)	
(c)	$1.8 + 0.4d - 0.002d^2 = -0.002(d^2 - 200d) + 1.8$	M1	1.1b
	$= -0.002((d - 100)^2 - 10000) + 1.8$	M1	1.1b
	$= 21.8 - 0.002(d - 100)^2$	A1	1.1b
		(3)	
(d)	(i) 22.1 metres	B1ft	3.4
	(ii) 100 metres	B1ft	3.4
		(2)	
(9 marks)			

Notes:
(a) M1: Sets $H = 0 \Rightarrow 1.8 + 0.4d - 0.002d^2 = 0$ M1: Solves using formula, which if stated must be correct, by completing square (look for $(d - 100)^2 = 10900 \Rightarrow d = \dots$) or even allow answers coming from a graphical calculator A1: Awrt 204 m only
(b) B1: States it is the initial height of the arrow above the ground. Do not allow "it is the height of the archer"
(c) M1: Score for taking out a common factor of -0.002 from at least the d^2 and d terms M1: For completing the square for their $(d^2 - 200d)$ term A1: $= 21.8 - 0.002(d - 100)^2$ or exact equivalent
(d) B1ft: For their '21.8+0.3' = 22.1m B1ft: For their 100m

Q10.

Question	Scheme	Marks	AOs
(a)	Sets $500 = \pi r^2 h$	B1	2.1
	Substitute $h = \frac{500}{\pi r^2}$ into $S = 2\pi r^2 + 2\pi r h = 2\pi r^2 + 2\pi r \times \frac{500}{\pi r^2}$	M1	2.1
	Simplifies to reach given answer $S = 2\pi r^2 + \frac{1000}{r}$ *	A1*	1.1b
		(3)	
(b)	Differentiates S with both indices correct in $\frac{dS}{dr}$	M1	3.4
	$\frac{dS}{dr} = 4\pi r - \frac{1000}{r^2}$	A1	1.1b
	Sets $\frac{dS}{dr} = 0$ and proceeds to $r^3 = k$, k is a constant	M1	2.1
	Radius = 4.30 cm	A1	1.1b
	Substitutes their $r = 4.30$ into $h = \frac{500}{\pi r^2} \Rightarrow$ Height = 8.60 cm	A1	1.1b
		(5)	
(c)	States a valid reason such as - The radius is too big for the size of our hands - If $r = 4.3$ cm and $h = 8.6$ cm the can is square in profile. All drinks cans are taller than they are wide - The radius is too big for us to drink from - They have different dimensions to other drinks cans and would be difficult to stack on shelves with other drinks cans	B1	3.2a
		(1)	
9 marks			

Notes:
(a) B1: Uses the correct volume formula with $V=500$. Accept $500 = \pi r^2 h$ M1: Substitutes $h = \frac{500}{\pi r^2}$ or $rh = \frac{500}{\pi r}$ into $S = 2\pi r^2 + 2\pi r h$ to get S as a function of r A1*: $S = 2\pi r^2 + \frac{1000}{r}$ Note that this is a given answer.
(b) M1: Differentiates the given S to reach $\frac{dS}{dr} = Ar \pm Br^{-2}$ A1: $\frac{dS}{dr} = 4\pi r - \frac{1000}{r^2}$ or exact equivalent M1: Sets $\frac{dS}{dr} = 0$ and proceeds to $r^3 = k$, k is a constant A1: $R =$ awrt 4.30cm A1: $H =$ awrt 8.60 cm
(c) B1: Any valid reason. See scheme for alternatives